

Developer UX, Information Architecture, and Contribution Research

Metadata	Details
Filename	02-GEMINI-DEV-UX-RESEARCH.md
Focus	Developer Experience (DevEx), Cognitive Interaction, System Architecture
Subject	dev.itisyou.app (Diagnostic Playbook Ecosystem)

Executive Summary

The landscape of software engineering is currently navigating a profound structural shift characterised by universal adoption of generative artificial intelligence, paired paradoxically with a collapse in developer trust regarding automated outputs. While 84% of developers in 2025 report using AI tools, trust in the accuracy of these tools has plummeted to 29%¹. The industry is experiencing a severe friction point: developers are relying on Large Language Models (LLMs) for speed, but are subsequently losing hours to debugging confidently incorrect, "almost right" AI-generated code¹. The stateless, forward-generation nature of these models makes them fundamentally unsuited for complex, multi-step diagnostic reasoning where maintaining an accurate mental model of system state is mandatory⁴.

Concurrently, traditional developer platforms are failing to fill this gap. Forums like Stack Overflow offer human verification but suffer from linear, unscalable question-and-answer formats that bury context and promote toxic gamification². The proposed platform, dev.itisyou.app, addresses this exact market failure. By transitioning the core interaction from a passive reading experience to an active, state-aware diagnostic playbook (Problem → Symptoms → Environment → Diagnostic Test → Result → Root Cause → Verified Fix), the platform introduces a deterministic, human-verified troubleshooting ecosystem. This report exhausts the human-computer interaction (HCI) requirements, cognitive load management strategies, information architecture, and contribution paradigms necessary to design this system. The analysis concludes that for the platform to succeed, it must aggressively utilize progressive disclosure, rely on structured micro-contributions to capture tacit knowledge, employ peer-verified reputation models to mitigate Sybil attacks, and utilize WAI-ARIA compliant interactive decision trees to guide developers from confusion to resolution.

Developer Behaviour Findings

Understanding the modern developer requires analyzing the cognitive mechanics of troubleshooting. Troubleshooting is a particularly intense problem-solving process that places sustained demands on a developer's attention, working memory, and mental modeling⁶. When an unexpected system behavior occurs, developers experience a spike in intrinsic cognitive load as they attempt to reconcile the system's actual state with their internal architectural model⁷. Prolonged exposure to this friction leads to severe cognitive fatigue⁶. When facing a broken state, developers currently fragment their attention across multiple inadequate sources. The decision of where to go is dictated by a subconscious calculation of required effort versus anticipated accuracy.

Troubleshooting Source	Primary Driver for Selection	Primary Failure Mode and Frustration Trigger
ChatGPT / Claude	Instantaneous response, syntactical generation, and contextual formatting ³ .	Hallucinated APIs, deprecated methods, and statelessness. The LLM cannot remember state beyond its context window, forcing the developer to repeatedly re-explain the broken environment ⁴ .
Stack Overflow	Search engine optimization dominance and the desire for human-verified, historical context ² .	Outdated answers dominating the top of threads, toxic moderation, and the requirement to read lengthy discussions to extract a single relevant terminal command ⁵ .
GitHub Issues	High trust. It is the closest source to the actual maintainers and raw codebase.	Endless scrolling through unformatted complaints. Issues are often closed as "stale" without a verified fix, leaving the developer stranded with a known but unsolved bug.
Official Documentation	Absolute trust regarding intended architecture and API definitions.	Documentation typically describes the "happy path" and rarely covers edge

		cases, environment-specific failures, or complex cross-framework integration issues.
Discord / Slack	Synchronous, immediate human-to-human interaction for highly niche or emerging frameworks.	Ephemeral knowledge. Solutions are buried in chat history, unsearchable via Google, and require the developer to wait asynchronously for a subject matter expert to reply.
Reddit / Blogs	Deep narrative explanations and architectural opinions.	Highly variable quality, rapid content decay, and a lack of standardized formatting.

Trust in a diagnostic tool is generated exclusively through deterministic accuracy and environmental parity. A developer trusts a solution if they can verify that the author experienced the exact same failure in the exact same software version. Frustration is generated by ambiguity, dead-ends, and the realization that the developer has invested cognitive effort into a solution that was never compatible with their environment to begin with. Developers tolerate very little effort before abandoning a tool; if a search result does not explicitly map to their error signature within seconds, they will bounce back to the search engine.

Entry Experience Findings

The lowest-friction starting user experience (UX) must account for the chaotic nature of software failure. Developers rarely arrive at a troubleshooting tool with a perfectly articulated natural-language question. They arrive with system exhaust.

The entry interface—ideally a prominent, omnivorous search input field—must be designed to ingest and gracefully parse raw, unformatted data. Developers should be able to paste a 50-line Java stack trace, a minified React error code, a cryptic Webpack terminal output, a Kubernetes pod log excerpt, or a standard natural-language query¹³.

The system must employ intelligent parsing (such as regex-based entity extraction and AI-assisted classification) to instantly identify the underlying technologies and the core error signature from this raw input¹⁶. If a developer pastes `TypeError: Cannot read properties of undefined (reading 'map')` along with a React component snippet, the UX should instantly filter the backend database for playbooks tagged with JavaScript, React, and array manipulation failures. Forcing the developer to manually select "Frontend" and "React" from a series of dropdown menus before searching introduces unnecessary extraneous cognitive load⁷ and

slows the time-to-resolution. The lowest-friction UX is a single, massive input field that acts as a universal diagnostic intake valve.

Core User Journeys

The foundational architecture of dev.itisyou.app relies on transitioning the developer from a state of high cognitive strain to a state of resolution through a strictly managed, state-aware sequence. The core user journey mirrors the cognitive phases of the "Theory of Troubleshooting"⁶.

The journey begins with **Symptom Ingestion**, where the developer inputs their broken state. This is immediately followed by **Environment Calibration**. The system cross-references the symptom with the user's specific runtime, framework, and operating system to dynamically prune irrelevant diagnostic branches, ensuring the developer is never presented with a Windows-specific fix on a Linux machine.

Once calibrated, the user enters **Stateful Diagnosis**. Unlike a forum that dumps all possible answers simultaneously, the system presents the highest-probability diagnostic test first. The developer executes the test in their local terminal or IDE and enters the **Feedback Loop**, reporting the outcome to the platform (e.g., "The command returned a 404"). Based on this explicit result, the journey proceeds to **Path Advancement**. The system's state-aware routing advances the developer to the next logical node in the decision tree, avoiding backtracking unless a dead-end is reached¹⁸.

Ultimately, the developer reaches the **Root Cause & Resolution** node, where the exact failure mode is identified and a verified fix is provided. The final step is **Confirmation**, wherein the developer clicks a low-friction toggle to confirm the fix worked, instantly generating a micro-contribution that strengthens the playbook's reputation data and assists future users.

Search UX Findings

Search is the primary entry vector and must serve as a rapid filtering mechanism. When a developer is triaging a production outage, they experience a high degree of time pressure. Presenting a search results page saturated with dense text, raw code snippets, and conversational prose creates an overwhelming visual hierarchy. The search results experience must prioritize scannability, allowing the developer to evaluate the relevance of a playbook in under three seconds.

Metadata Element	UX Recommendation	Rationale
Problem Title	Essential & Prominent	Must be written in clear, symptom-first language (e.g., "Kubernetes Pod CrashLoopBackOff due to OOMKilled").

Technologies & Versions	Essential	Displayed as high-contrast visual tags. If a developer is using Node 22, seeing a "Node 14" tag instantly saves them from pursuing a deprecated path.
Verification Status	Essential	A trust badge (e.g., "Community Verified" or "Maintainer Approved") serves as the primary heuristic for quality ² .
Estimated Diagnostic Time	Highly Useful	Setting an expectation (e.g., "5 mins to diagnose") respects the developer's urgency and reduces anxiety regarding the depth of the rabbit hole.
Last Verified Date	Highly Useful	In modern web development, a fix from 2022 is often a liability in 2026. Recency indicates validity.
Reproduction Count	Useful (Aggregated)	Showing "1,200 developers fixed this" builds empirical trust, but granular breakdowns should be hidden on the search page.
Difficulty	Marginal Utility	"Difficulty" is highly subjective. A senior systems engineer and a junior frontend developer will perceive the same regex issue entirely differently.

Success/Failure Evidence	Overwhelming	Displaying raw CI/CD logs or detailed success matrices in the search view consumes too much vertical space and prevents rapid scanning.
Raw Code Snippets	Overwhelming	Code should only be shown if it is the exact search match; otherwise, it clutters the interface and distracts from the architectural context.

Playbook UX Findings

The interface for the troubleshooting playbook is the core distinguishing feature of dev.itisyou.app. HCI research comparing various user interface patterns reveals distinct failure modes for traditional approaches when applied to complex diagnostics.

Long articles suffer from linear constraints; they cannot elegantly handle conditional logic (e.g., "If condition A, scroll to section 4; if condition B, scroll to section 8"). This forces the developer to hold the routing logic in their working memory⁶. Checklists are too simplistic and assume a single, unbroken path to success. Steppers and command-by-command guides artificially constrain visibility, preventing developers from anticipating future steps or understanding the overall architecture of the problem¹⁸. Conversational assistants obscure the overarching structure entirely, generating a "monologue" rather than a state-aware diagnostic map¹⁸.

The optimal pattern is a **State-Aware Split View**. The left panel serves as "The Map," displaying the interactive decision tree or flowchart. It highlights the current active node, the history of completed nodes, and the immediately available branches. This provides absolute clarity and serves as an architectural blueprint for the debugging session¹⁸. The right panel serves as "The Territory," displaying the dense contextual information for the currently active node selected on the left. This includes the specific CLI commands, code blocks, environmental warnings, and expected outputs.

This split-view architecture allows the developer to move seamlessly through the Test → Result → Next Action sequence. They can read the deep technical context on the right, execute the command, and click their result on the left to advance the tree, all without ever losing their place in the broader diagnostic context.

Diagnostic Tree UX Findings

Visualizing complex, non-linear diagnostic trees introduces significant user interface challenges. A software decision tree must clearly define pivotal moments (nodes), user actions (branches), and terminal points (leaves) without overwhelming the user¹⁸. Real-world

diagnostic trees—such as those used to diagnose vector database latency or email authentication failures—are often multi-layered and heavily branch based on environmental constraints²¹.

Visual Challenge	Recommended UX Presentation
Branches & Deep Trees	Implement aggressive progressive disclosure. Displaying a 50-node tree simultaneously induces cognitive paralysis. The UI should highlight the active path in full opacity, display immediate child branches, and hide deep descendants until their parent condition is explicitly met.
Uncertainty & Optional Steps	Diagnostics are rarely perfectly binary. The tree must support "Unknown," "Skip," or "Output Unclear" branches. Selecting these should gracefully route the user to fallback heuristic tests or broader architectural checks, rather than forcing a hard failure.
Multiple Root Causes	If a symptom maps to multiple potential root causes (e.g., a 502 Bad Gateway), the tree should visually group the initial diagnostic branches by probability or frequency of occurrence, guiding the user toward the most likely culprit first.
Environment-Specific Paths	Branches that are fundamentally incompatible with the user's declared environment (e.g., a Windows registry fix when the user is on Ubuntu) should be visually collapsed, struck through, or dimmed. They must remain discoverable to prevent confusion, but should not command primary visual attention.
Backtracking	Developers frequently execute a test incorrectly or mistype a command. The UI must allow users to click any previously completed node, invalidate the subsequent

	path, and select an alternate branch without resetting the entire playbook state.
Failed vs. Completed Tests	Use standardized semantic colouring. Completed nodes should feature muted success indicators (e.g., subtle green borders), while failed or abandoned nodes should utilize warning aesthetics, allowing the developer to visually retrace their diagnostic history at a glance.

Knowledge Density Findings

Developer content possesses an inherently high knowledge density. A single diagnostic node might require the presentation of a shell command, a JSON configuration file, a stack trace, and a theoretical explanation of the underlying protocol. Exposing all this information simultaneously violates the principles of managing extraneous cognitive load⁷.

To manage this density, the right-hand context panel must leverage progressive disclosure rigorously. Immediately visible elements must include the primary action (the exact code or CLI command to run), the expected result (what the terminal should output if the system is healthy), and any critical warnings. Destructive actions, such as dropping a database table or flushing a cache, must utilize high-contrast, universally understood warning patterns that interrupt the user's workflow before execution.

Elements that should be hidden behind expandable accordions or tabs include raw log files, lengthy stack traces, deep theoretical explanations, and historical context.

Code presentation requires specialized, flawless UX. Syntax highlighting is non-negotiable. Code blocks must include one-click "Copy to Clipboard" functionality. Furthermore, there must be a distinct visual separation between input commands and expected output. A standard convention is prefixing commands with a \$ and rendering output text in a muted, non-selectable font. Additionally, environmental variables that require user-specific substitution (e.g., <YOUR_API_KEY>) must be visually distinct, perhaps utilizing a highlighted background, to prevent users from blindly copying placeholder text into their terminals.

Contribution UX Findings

The long-term viability and scalability of dev.itisyou.app rests entirely on its contribution model. Traditional documentation requires high activation energy; developers must write prose, format markdown, and structure complex logic. This high friction deters participation, resulting in the persistent loss of tacit troubleshooting knowledge on the engineering floor²³.

To minimize friction, the platform must facilitate decentralized, structured micro-contributions combined with central, automated curation²⁴. The system must abandon the expectation that developers will author formal technical articles.

Evaluating the proposed contribution approaches reveals stark differences in friction:

- **Paste Rough Notes and AI Structures Them:** This is the optimal, lowest-friction method. A developer can dump a raw stream of consciousness, a terminal history, and an unformatted stack trace into an ingestion field. An integrated LLM—acting strictly as a structural parser, not a generator—translates this raw input into the platform's rigid node-branch-leaf JSON format²⁶. The developer then acts merely as an editor and reviewer, confirming the AI's structural accuracy rather than authoring from a blank slate.
- **Add One Missing Diagnostic Test:** Essential for iterative improvement. If a developer finds that a playbook works, but requires an extra chmod step on their specific OS, they must be able to branch the tree with a micro-contribution directly from the node interface.
- **Convert Solved Issue:** Deep integration with source control platforms (GitHub/GitLab) to automatically ingest closed issues and their verified pull requests, using AI to translate the discussion thread into a diagnostic path, requires zero manual drafting from the developer.
- **Report "This Worked" / "This Failed":** The absolute lowest form of micro-contribution, requiring only a single click to provide empirical validation data to the system.

Conversely, approaches like "Guided structured forms" or "Import Markdown" require developers to manually adapt to the platform's proprietary syntax or spend time filling out extensive fields, drastically increasing the likelihood of form abandonment²⁷.

"This Worked" Reproduction UX

The lightest form of micro-contribution is the reproduction report. This telemetry is critical for generating empirical trust data and mapping version compatibility. Because this action occurs at the very end of the developer journey, when the developer is eager to return to their primary coding task, the flow must take no more than 10 to 30 seconds to complete. Exceeding this time budget will cause form abandonment to spike²⁷.

Data Field	Friction Level	UX Implementation Strategy
Worked / Partial / Failed	Minimal	A simple, prominent trinary toggle presented immediately upon reaching a terminal "Fix" node.
OS, Framework, Runtime	Low (if inferred)	If the user is authenticated and has a profile, or used specific search filters, these should be pre-populated. The user only needs to click "Confirm." If manual entry is

		required, it should use highly optimized autocomplete dropdowns.
Environment Details	Moderate	Only request broad categories (e.g., "Local," "Staging," "Production"). Do not ask for deep hardware specs unless the playbook is explicitly tagged as hardware-dependent.
Root Cause / Notes	High	Must be entirely optional. A single text area for "Notes or missing steps." Forcing a developer to write a justification here guarantees they will close the tab.

The primary goal of this flow is statistical volume. The system relies on aggregating thousands of low-friction signals to build a reliable, irrefutable compatibility matrix.

Reputation UX

Gamification models based solely on raw upvotes, view counts, or generic points—as utilized by traditional forums—invariably deteriorate into popularity contests, encourage low-effort spam, and foster toxic, elitist gatekeeping⁵.

A diagnostic playbook ecosystem requires a reputation system rooted strictly in peer verification, empirical utility, and anti-abuse mechanisms¹⁹. The goal is to mathematically identify and reward users who provide accurate, deterministic solutions that survive contact with the real world.

Reputation Signal	Value	Systemic Impact and Risk Mitigation
Successful Reproductions	Highest	The ultimate metric of truth. If an author publishes a playbook and 500 independent developers

		confirm it resolved their issue, the author's reputation scales proportionately.
Maintainer Confirmation	Very High	An explicit endorsement from a core maintainer of the technology (e.g., a Rust core team member verifying a Cargo playbook). Acts as a massive multiplier for trust.
Specialist Technology Areas	High	Reputation must be faceted. A user might possess a "Verified Expert" status in PostgreSQL diagnostics, but zero reputation in React. Global reputation scores misrepresent actual competence.
Playbook Success Rate	High	The ratio of successful to failed reproductions. A developer who publishes one flawless playbook is more valuable than one who publishes fifty playbooks with a 20% success rate.
Speed of Contribution	Dangerous	Rewarding the "first to answer" incentivizes users to paste unverified, LLM-generated slop just to claim the real estate.
Generic Upvotes	Dangerous	Highly susceptible to Sybil attacks, where malicious actors script fake identities

		to artificially inflate the credibility of insecure or malicious playbooks ²⁸ .
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Verification and Trust UX

Visualizing trust is critical because developers are currently suffering from a severe crisis of confidence regarding the accuracy of their tools¹¹. The interface must communicate reliability instantly, without resorting to "badge overload," which creates visual noise and diminishes the semantic impact of the badges themselves.

The trust hierarchy should be communicated through a standardized, strictly controlled visual language:

1. **Gold Standard (Human-Verified / Officially Supported / Maintainer Confirmed):** Indicates the playbook has been tested by the community with a statistically significant success rate, or explicitly verified by the technology's creators. Visualized with distinct, high-contrast, universally positive iconography (e.g., a solid green shield or a definitive checkmark).
2. **Silver Standard (Machine-Tested / AI Reviewed):** Indicates the core code blocks and configurations have been statically analyzed or executed in a sandboxed CI environment to ensure they do not throw immediate syntax errors, but the playbook lacks empirical human testing. Visualized with neutral, structural icons (e.g., a grey terminal icon or a cog).
3. **Warning State (Unverified / Needs Reverification / Deprecated):** Indicates a newly contributed playbook lacking reproductions, or an older playbook utilizing methods that have since been deprecated by the upstream framework. Visualized with cautionary aesthetics (e.g., amber warning triangles or dashed borders).

Explicitly separating "AI Generated" from "Human Verified" is essential. As AI-generated content permeates the web, human verification becomes the platform's premium commodity².

Versioning and Compatibility UX

Software troubleshooting is entirely dependent on environment parity. A solution that works perfectly in Node 22 may catastrophically fail in Node 26. Simple text labels (e.g., "Requires Ubuntu") are insufficient for representing complex, multi-variable software stacks.

The most intuitive and scalable way to display this data is through a **Compatibility Matrix**, a design pattern established and perfected by the Mozilla Developer Network (MDN) Web Docs for browser compatibility data³⁰.

The matrix should visualize the intersection of major variables (e.g., Operating System on the Y-axis, Framework Version on the X-axis). The cells within the matrix should utilize clear, unmistakable indicators:

- **Green Check:** Explicitly verified to work, backed entirely by user "This Worked" reports.
- **Red Cross:** Explicitly verified to fail.
- **Grey Question Mark:** Unknown or untested. Crucially, this state serves as an interactive call-to-action, explicitly inviting micro-contributions ("Are you on Windows 11? Click here

to report if this works")³⁵.

This matrix must be prominently displayed at the very beginning of the playbook experience, allowing developers to instantly determine if the diagnostic tree applies to their stack before they invest any cognitive effort into reading it.

Accessibility Findings

Given the platform's heavy reliance on complex visual structures, split views, and interactive nodes, robust accessibility is paramount. Custom interactive components must adhere strictly to W3C WAI-ARIA (Web Accessibility Initiative - Accessible Rich Internet Applications) standards³⁶. Without these, the core value proposition of the site is completely walled off from developers relying on assistive technologies.

The interactive diagnostic tree must utilize the ARIA tree or treegrid roles³⁷. Individual nodes must use treeitem roles, and their state must be explicitly managed with the aria-expanded property (true/false) to indicate to screen readers whether child branches are currently visible³⁷.

Keyboard navigation is critical, as power users and developers frequently rely on keyboards over mice. The tree must support standard directional arrow-key navigation³⁹. Pressing the 'Enter' or 'Space' key should activate a node and dynamically update the right-hand context panel⁴⁰. Focus states must be highly visible, utilizing high-contrast borders that do not rely on subtle color shifts.

If SVG elements are used to render visual flowcharts or paths between nodes, the SVGs must include semantic grouping and descriptive aria-labels to explain the flow. Relying purely on visual lines to denote relationships will alienate visually impaired developers⁴². Furthermore, terminal output and status indicators (success, failure, compiling) must not rely on color alone; red error text must be accompanied by an icon or a text prefix (e.g., [ERROR]). Finally, any transitions between tree nodes or panel slides must respect the operating system's prefers-reduced-motion media query to accommodate users with vestibular disorders.

Mobile Findings

Serious software troubleshooting is a desktop-heavy activity. Developers do not write deep architectural patches, parse 500-line log files, or execute CLI diagnostics from a smartphone. However, mobile usage in the developer lifecycle occupies a specific, vital niche: triage, asynchronous reading, and review. A developer might receive a Slack alert about a failing pipeline while commuting, and use their phone to read a playbook to understand the architectural implications before they reach their laptop.

Therefore, the mobile UX should not attempt to replicate the interactive split-view desktop troubleshooting environment. Doing so would result in a cramped, unusable interface. Instead, the mobile UX should degrade gracefully into a linear, read-only "Map" of the diagnostic tree. Interactive node testing should be visually de-emphasized. Code blocks must support horizontal scrolling without breaking the viewport or triggering unintended vertical scrolls. The primary mobile call-to-action should shift from "Execute this command" toward asynchronous actions like "Save to Profile for Later," "Copy URL," or "Send to Slack."

Information Architecture Findings

The top-level information architecture must reflect the urgency of the user's intent. When production infrastructure is failing, developers do not want to navigate through deeply nested category trees or browse curated content feeds.

A **Search-First Architecture** is unequivocally stronger for this product. The interface should prominently feature a highly capable, omni-present search bar capable of parsing natural language, exact error codes, and log excerpts¹³.

Top-level navigation should be minimal, utility-focused, and state-driven:

1. **Search (Omnibox):** The core routing mechanism, front and center.
2. **My Environment:** A persistent user state where a developer defines their current stack (e.g., AWS, Kubernetes, React, Node 22). When set, this automatically filters all future searches and prunes incompatible branches from trees.
3. **Playbooks (Browse):** A secondary path for discovery, sorted strictly by verified status and technology, serving developers who are researching best practices proactively rather than reactively.
4. **Contribute:** Quick, global access to the AI-assisted rough-notes parser.
5. **Profile / Reputation:** A dashboard tracking the user's micro-contributions, peer verifications, and saved states.

Traditional categories (e.g., "Databases", "Frontend", "Networking") should exist as facets and filters *after* a search is executed, rather than serving as the primary navigation portals on the homepage.

Homepage Experience

The five-second test is critical for the homepage. When a developer lands on dev.itisyou.app, they must immediately understand the platform's unique value proposition.

The site must explicitly distance itself visually and functionally from three archetypes:

- **Another Social Network:** It must not feature an infinite feed of opinions, trending topics, or highly visible user avatars dominating the hierarchy.
- **Another AI Chatbot:** It must not present a blank chat window with a generic "How can I help you today?" prompt. Developers associate this with probabilistic, hallucination-prone interfaces⁴.
- **Another Documentation Site:** It must not look like a static wiki or a wall of Markdown text.

The homepage must communicate structure, determinism, and action. The visual focal point should be the massive, omnivorous diagnostic search bar, paired with a sub-header explicitly stating the platform's premise: "Paste your error. Follow the verified diagnostic tree. Fix the root cause." Below the fold, the homepage should display a live, rotating example of a split-view diagnostic tree to immediately demonstrate the interactive, state-aware nature of the product, proving that this is not a forum, but a living playbook.

Community UX Risks

The design of the community interactions carries severe risks that could destroy the platform's credibility if mismanaged.

With the ease of LLM generation, bad actors will attempt to flood the platform with auto-generated, untested playbooks to farm reputation. The system must sandbox untested contributions, applying the "Unverified" trust badge, and rely entirely on the "Peer Verification" signal before elevating content in the search index¹⁹. Platforms that allow high-reputation users to unilaterally close, lock, or edit threads often breed intense resentment and "toxic moderation" cultures⁵. Dev.ITISYOU should avoid subjective editorial locking. If a playbook is deemed ineffective by the community, it should naturally sink in the search rankings due to a low mathematical "Success Rate" metric, rather than relying on the subjective decisions of human gatekeepers. Finally, the platform must strictly forbid open-ended forum threads. Endless discussions do not solve bugs. If a user disagrees with a diagnostic step, the interface should funnel them into proposing a definitive *branch* or a new *heuristic test*, rather than allowing them to leave a 500-word argumentative comment. Discourse must be forced into the structure of the tree.

Stitch Design Requirements

To ensure the subsequent UI design phase via Google Stitch is structurally sound, the following constraints, components, and requirements must be strictly adhered to:

Requirement Category	Specific Directives
Required Page Types	1. The Omnibox / Landing: Focused entirely on accepting raw input (logs, errors, queries). 2. Search Results / Triage: Displaying filtered lists with compatibility matrices and trust badges. 3. The Playbook (Split-View): Left-side interactive tree component; Right-side deep context panel. 4. Contribution Pipeline: Multi-step flow for raw input ingestion, AI structuring review, and publication. 5. Profile / Reputation: Dashboard tracking reproductions and trust tier.
Key Components	1. Interactive Tree Node: Must support distinct visual states (locked, active, completed, failed, unknown).

	2. Compatibility Matrix Table: Visual grid for OS/Framework/Version intersections based on the MDN pattern. 3. Code Block: With one-click copy, syntax highlighting, and visual separation of output vs. input. 4. Trust Badge: Standardized iconography for Verified, Machine-Tested, and Unverified.
Important States	1. Backtracking State: Visual indication when a user invalidates a previous node and selects a new branch. 2. Environment Mismatch: Clear visual dimming or warning when a user views a playbook incompatible with their declared profile stack.
Responsive Considerations	The split-view Playbook must collapse into a linear, stacked layout on viewports narrower than 1024px. The interactive tree must become scrollable or simplified on mobile devices to prevent horizontal overflow.
Accessibility Requirements	Strict adherence to WAI-ARIA treegrid implementation. All interactive elements must have visible keyboard focus rings (minimum 2px solid border) and maintain minimum 4.5:1 color contrast ratios³⁶.
Design Pitfalls	Avoid embedding critical navigation within the tree itself; always provide escape hatches. Do not use color alone to convey node state (success/failure). Do not clutter the right-hand panel with unformatted text.

UX Principles	Enforce progressive disclosure. Prioritize data density over whitespace in technical panels, but maintain clear typographical hierarchy.
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Open UX Questions

Several interactions require rapid prototyping and A/B testing with active developers to validate assumptions:

1. **AI Ingestion Accuracy Limits:** When a user pastes a highly fragmented, messy terminal history, how accurately can the structural LLM parse it into a deterministic diagnostic tree without hallucinating missing intermediary steps?
2. **Tree Density Thresholds:** At what specific node count does the visual representation of the decision tree overwhelm the user, requiring forced pagination, zooming, or sub-tree collapsing?
3. **Authentication Friction:** Can the platform gather meaningful, Sybil-resistant "This Worked" telemetry from unauthenticated users via browser fingerprinting or session tracking without violating privacy norms or corrupting the reputation system?

Recommended UX Principles

1. **State Over Static:** Always maintain and visualize the user's context. A diagnostic process is an ongoing conversation with the system, not a static document. The UI must remember where the developer is and what they have already tried.
2. **Progressive Disclosure Over Density:** Expose only the information necessary for the immediate decision. Hide raw logs, deep theory, and complex matrices until they are explicitly required or requested by the user.
3. **Human Verified Over AI Generated:** In an era defined by infinite synthetic content, the most valuable signal is a human stating, "I ran this, and it worked." Design every interaction to capture, elevate, and celebrate that empirical signal.
4. **Deterministic Over Probabilistic:** Avoid chat interfaces that generate unique, unpredictable, and unrepeatable answers. Present concrete, repeatable, and testable paths that guarantee a specific outcome based on specific inputs.
5. **Zero-Friction Ingestion:** Accept developer inputs in their rawest forms—messy stack traces, unformatted logs, and broken syntax. Let the system do the heavy computational lifting of parsing, formatting, and structuring.

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